Bureau
International des
Poids et
Mesures

Le système international d'unités
The International System of Units

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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Aluminium 27 Ion ()

Aluminium 27 Ion ()

CCL-CCTF Frequency Standards Working Group



Bureau International
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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 2: Aluminium 27 Ion ()

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Appendix 2

1. Recommended value [1] of the frequency

$f\left(^{27}\mathrm{Al}^{+}\right) = 1\ 121\ 015\ 393\ 207\ 857.3\ \mathrm{Hz}$

equivalent to

$\lambda\left(^{27}\mathrm{Al}^{+}\right) = 267\ 429\ 385.730\ 489\ \mathrm{fm},$

with an estimated relative standard uncertainty of 1.9×10^{-15}.

This radiation was endorsed as a secondary representation of the definition of the second.

2. Source data

$f\left(^{27}\mathrm{Al}^{+}\right) = 1\ 121\ 015\ 393\ 207\ 857.3\ \mathrm{Hz}$

$u_c/y = 1.9 \times 10^{-15}$

calculated from

$f\left(^{27}\mathrm{Al}\right)$/ Hz	u/ Hz	source data
1 121 015 393 207 851	6	[2]
1 121 015 393 207 857.4	0.7	[3]

by a weighted mean.

As this value was issued from only one laboratory, the CCTF considered it prudent to estimate a relative standard uncertainty of 1.9×10^{-15}.

$^{27}Al^{+}$ ion, $3s^2\ {}^1S_0 - 3s3p\ {}^3P_0$ unperturbed optical transition

References

[1] CIPM Recommendation 1(CI-2013): Updates to the list of standard frequencies.

[2] T. Rosenband, P. O. Schmidt, D. B. Hume, W. M. Itano, T. M. Fortier, J. E. Stalnaker, K. Kim, S. A. Diddams, J. C. J. Koelemeij, J. C. Bergquist, and D. J. Wineland, Observation of the $^{1}S_{0} \rightarrow {}^{3}P_{0}$ Clock Transition in $^{27}Al^{+}$, *Phys. Rev. Lett.* **98**, 220801 (2007).

[3] T. Rosenband, D. B. Hume, P. O. Schmidt, C. W. Chou, A. Brusch, L. Lorini, W. H. Oskay, R.E. Drullinger, T. M. Fortier, J. E. Stalnaker, S. A. Diddams, W. C. Swann, N. R. Newbury, W. M. Itano, D. J. Wineland and J. C. Bergquist: Frequency Ratio of Al^{+} and Hg^{+} Single-Ion Optical Clocks; Metrology at the 17th Decimal Place, *Science* **319**, 1808 (2008).